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Question Paper Code : 15314

B.E. / B.TECH. DEGREE EXAMINATION, APRIL / MAY 2019

Third Semester

B.E. – Computer Science and Engineering

15CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. What is the use of Condition Code register?

2. Using Booth's algorithm for multiplication, the multiplier -57 will be re-coded as

   A) 0 -1 0 0 1 0 0 -1 B) 1 1 0 0 0 1 1 1

   C) 0 -1 0 0 1 0 0 0 D) 0 1 0 0 -1 0 0 1

3. List the different stages involved in instruction execution.

4. A CPU has 24-bit instructions. A program starts at address 300 (in decimal). Which one of the following is a legal program counter (all values are in decimal)?

   A) 400 B) 500 C) 600 D) 700

5. Register renaming is done in pipelined processors

   A) as an alternative to register allocation at compile time

   B) for efficient access to function parameters and local variables

   C) to handle certain kinds of hazards

   D) as part of address translation

6. A 5-stage pipelined processor has Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (E) and Write Operand (W) stages. The IF, ID, OF and W stages take 1 clock cycle each for any instruction. The E stage takes 1 clock cycle for ADD and SUB instructions, 3 clock cycles for MUL instruction and 6 clock cycles for DIV instruction respectively. Operand forwarding is used in the pipeline.

What is the number of clock cycles needed to execute the following sequence of instructions?

Instruction	Meaning of instruction
I0 :MUL R2, R0, R1	R2 = R0*R1
I1 :DIV R5, R3, R4	R5 = R3/R4
I2 :ADD R2, R5, R2	R2 = R5+R2
I3 :SUB R5, R2, R6	R5 = R2-R6

A) 13 B) 15 C) 17 D) 19

7. What is the purpose of dirty / modify bit?

8. How many 32K x 1 RAM chips are needed to provide a memory capacity of 256K-bytes?

A) 8 B) 32 C) 64 D) 128

9. Distinguish between memory-mapped and program-controlled I/O.

10. What is the purpose of DOUT bit in display device register?

PART B – (5 × 16 = 80 Marks)

11. a) i. Give a brief account on indexed and indirect addressing modes with appropriate examples. (4 Marks)

11. a) ii. Using Booth's algorithm, perform the operation 13 x (-7) (6 Marks)

11. a) iii. Using Sequential Circuit Multiplier, perform 14 x 9 (6 Marks)

OR

11. b) i. Write notes on Instruction sequencing with an appropriate example. (4 Marks)

11. b) ii. Multiply the following pair of signed 2's complement numbers using bit-pair recoding of the multipliers: A=1011, B=1100 (6 Marks)

11. b) iii. Perform the division on the following 4-bit unsigned integers using restoring division: 1011 / 0111. (6 Marks)

12. a) i. Explain with the neat diagram, the control signals used in Processor Memory Interface, Instruction Register and Instruction Address Generator of a RISC style processor. (8 Marks)

12. a) ii. At the time the instruction Load R6, 100(R7) is fetched, R6 and R7 contain the values 4200 and 5320 respectively. Memory location 4300 and 5420 contain 50 and 75 respectively. Write the Register transfer notation and show the contents of the inter-stage registers in data path during each of the 5 execution steps of this instruction. (8 Marks)

OR

12. b) i. Explain with the neat diagram, the data path of a RISC style processor. (8 Marks)

12. b) ii. List the sequence of actions needed to execute the following instructions in terms of data path components:

A) STORE R5, 10(R6)

B) SUBTRACT R3, R4, R5 (8 Marks)

13. a) i. With the support of the modified data path of a RISC style processor, discuss how operand forwarding is performed to overcome pipelining hazards. (8 Marks)

13. a) ii. Consider the following instructions at the given addresses in the memory:

1000 ADD R3, R2, #100

1004 SUBTRACT R5, R4, #25

1008 AND R6, R4, #0X37

1012 ADD R7, R2, R4

Initially, registers R2 and R4 contain 1500 and 500, respectively. These instructions are executed in a computer that has a four-stage pipeline. The first instruction is fetched in clock cycle 1, and the remaining instructions are fetched in successive cycles. With reference to data path, identify the contents of inter stage buffers for clock cycles 2 to 6. (8 Marks)

OR

13. b) i. Discuss dynamic branch prediction. Also explain how branch target buffer is used in dynamic branch prediction. (8 Marks)

13. b) ii. Assume that 40 percent of the dynamic count of the instructions executed for a program is branch instructions. There are no pipeline stalls due to data dependencies.

A) Determine the execution times for two cases: when 35 percent of the branches are taken, and when 65 percent of the branches are taken.

B) Determine the speedup for one case relative to the other. Express the speedup as a percentage relative to 1.01. (8 Marks)

14. a) i. Design a 512K x 4 memory chip module. Also provide the neat sketch of the same. (8 Marks)

14. a) ii. A block-set-associative cache consists of a total of 512 blocks divided into 8-block sets. The main memory contains 16384 blocks, each consisting of 256 words.

A) How many bits are there in a main memory address?

B) How many bits are there in each of the TAG, SET and WORD fields? (8 Marks)

OR

14. b) i. Consider a 2 level cache with access time of 5ns and 80ns and hit rates 82% and 93% respectively and memory access time is 250ns. What is the average access time? (8 Marks)

14. b) ii. Suppose that a computer has a processor with two L1 caches, one for instructions and one for data, and an L2 cache. Let τ be the access time for the two L1 caches. The miss penalties are approximately 15τ for transferring a block from L2 to L1, and 100τ for transferring a block from the main memory to L2. For the purpose of this problem, assume that the hit rates are the same for instructions and data and that the hit rates in the L1 and L2 caches are 0.96 and 0.80, respectively.

A) What fraction of accesses misses in both the L1 and L2 caches, thus requiring access to the main memory?

B) What is the average access time as seen by the processor?

C) Suppose that the L2 cache has an ideal hit rate of 1. By what factor would this reduce the average memory access time as seen by the processor? (8 Marks)

15. a) i. Explain the functionality of asynchronous bus with a neat sketch. (8 Marks)

15. a) ii. With appropriate diagrams, discuss PCI bus structure and also explain the different data transfer signals associated with PCI bus. (8 Marks)

OR

15. b) i. Explain how IO devices can be interfaced, with a neat diagram. (8 Marks)

15. b) ii. With a neat sketch of different registers for keyboard, explain the routine for input process. (8 Marks)